TEB2164 Introduction to Data Science
Lab 9b- Correlation Analyses and Normalization in R

Learning Outcomes

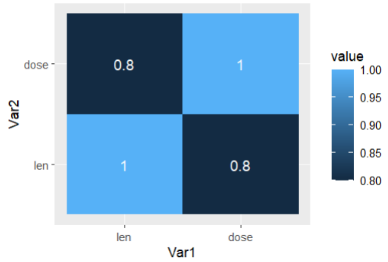
At the end of the session, you will be able to:

- Understanding correlation
- Plotting correlation with heatmap
- Understanding normalization with log transformation, standard scaling and min-max scaling

Activity

1. Using “ToothGrowth” dataset, compute correlation analysis and plot the correlation using a heatmap. Discuss your observation/insight based on the findings.

```
library(lattice)
data = ToothGrowth[, c(1,3)]
corr_mat <- round(cor(data),2)
dist <- as.dist((1-corr_mat)/2)
hc <- hclust(dist)
corr_mat <-corr_mat[hc$order, hc$order]
library(reshape2)
melted_corr_mat <- melt(corr_mat)
library(ggplot2)
ggplot(data = melted_corr_mat, aes(x=Var1, y=Var2,fill=value)) +
geom_tile() + geom_text(aes(Var2, Var1, label = value), color = "white",
size = 4)
```



The tooth length is 0.8 correlated to the dose of vitamin C given per day. Any other logical observation is acceptable.

2. Using “mtcars” dataset, compute normalization using log transformation, standard scaling and min-max scaling. Compare and contrast your finding of the three results. Discuss your findings.

```
scaled_data1 <- log(mtcars)
print(scaled_data1)
scaled_data2<- as.data.frame(scale(mtcars))
print(scaled_data2)
minmax <- preProcess(as.data.frame(mtcars), method=c("range"))
scaled_data3 <- predict(minmax, as.data.frame(mtcars))
print(scaled_data3)
summary(mtcars)
summary(scaled_data1)
summary(scaled_data2)
summary(scaled_data3)
```

Submission

- Submit to your GA.